

Problem Set 2

1. Sinusoid through an LTI system

An LTI system has $H(e^{j\omega}) = 1 + 0.5e^{-j\omega}$. The input is $x[n] = 2\cos(\pi n/3 + \pi/4)$.

- Evaluate $H(e^{j\pi/3})$ in rectangular form.
- Find its magnitude and phase.
- Write the steady-state output $y[n]$ in cosine form.

2. DTFT shifting and a spectral zero

Let $x[n] = (0.5)^n u[n]$ and $y[n] = x[n - 2] - x[n]$.

- Find $X(e^{j\omega})$.
- Use the time-shift property to find $Y(e^{j\omega})$.
- Evaluate $Y(e^{j0})$ and explain what that value says about the sum of $y[n]$.

3. Difference equation to frequency response

A causal system satisfies $y[n] - 0.8y[n - 1] = x[n] - x[n - 1]$, with zero initial conditions.

- Derive $H(e^{j\omega})$.
- Evaluate H at $\omega = 0$ and $\omega = \pi$.
- Based only on those values and the structure of the numerator, is the system more “low-pass-like” or “high-pass-like”? Explain.
- Use `scipy.signal.freqz` to check your reasoning.

4. From continuous frequency to discrete frequency

A microphone voltage is $x_c(t) = 2\cos(2\pi \cdot 1000 t) + \cos(2\pi \cdot 5000 t)$. It is sampled at $f_s = 16$ kHz.

- Write $x[n]$ explicitly.
- Give the two discrete-time angular frequencies in rad/sample and as multiples of π .
- Are either components aliased at this sample rate? Explain.

5. Spectral replicas and Nyquist margin

An analog signal is strictly bandlimited to $|f| \leq 3.4$ kHz and is sampled at $f_s = 8$ kHz.

- a) Sketch or describe the locations of the baseband spectrum and the first two pairs of replicas after ideal sampling.
- b) What is the separation between adjacent spectral replicas?
- c) What guard-band width exists between the positive edge of the baseband copy and the nearest edge of the $+f_s$ replica?
- d) What is the ideal Nyquist rate for this signal? Is 8 kHz above it?

6. Downsampling audio: Filter first

A recording is sampled at 48 kHz and contains useful content up to 18 kHz. You want a 16 kHz output by downsampling by 3.

- a) What is the new Nyquist frequency?
- b) Why is keeping every third sample without filtering incorrect?
- c) What band must be strongly attenuated before downsampling?
- d) If an 11 kHz sinusoid is present and you downsample naively to 16 kHz, at what frequency does it appear?

7. Prove two analog sinusoids give the same samples (ECEN 5632 problem)

At $f_s = 12$ kHz, compare $x_1(t) = \cos(2\pi \cdot 2000t)$ and $x_2(t) = \cos(2\pi \cdot 10000t)$.

- a) Write $x_1[n]$ and $x_2[n]$.
- b) Use a trigonometric or complex-exponential identity to prove $x_1[n] = x_2[n]$ for every integer n .
- c) Explain in one sentence why this proves aliasing cannot generally be undone after sampling.